

Technical Support Workbook Answer Sheet

Name:

Date:

Session 1: The Foundations of IT Support and Hardware Basics

1.B	2.C	3.B	4.C	5.B
6. Speak directly and in a polite manner. Be a good listener and an empathizer. Follow up to see that the problem is completely solved.				
7. Is the computer plugged and is the power switch on?				
8. HDD: Rotating magnetic disks are used to store data (mechanical, slower). SSD: Flash memory is used (no moving components, faster, more durable).				
9. • CPU socket • RAM slots				
10. Pausing until the user gives some information (e.g., serial number, or approval or testing confirmation).				
Research Task: Quantum computing refers to the latest type of computing, which involves the use of the weird laws of quantum physics to solve some problems significantly quicker than conventional computers. With a quantum computer, in contrast to a regular computer, which uses bits (0 or 1), a quantum computer uses so-called qubits, which can be in 0 and 1 simultaneously - a phenomenon known as superposition. It also employs an effect known as entanglement whereby qubits can be connected to each other in a way where the state of one will be instantaneously changed by the states of others. Due to these characteristics, quantum computers may address some highly complex problems, such as simulating large molecules in drug development, breaking some encryptions or optimising supply chains - problems that classical computers cannot easily solve fast. Nonetheless, the technology is in its infancy: qubits are highly delicate, and they require extreme cold or thorough isolation, and error-correction is a significant challenge. Some researchers reckon that it can be several years before a useful, large-scale quantum computer is built, although recent discoveries have indicated that the timer is about to run out. To the average user it is to say that in the future we are likely to have devices or services that are run using quantum computers that can provide new medicines with increased speed, have better logistics or enhance AI - however at the moment normal computing is quite sufficient. References National Institute of Standards and Technology. (2025). <i>Quantum computing explained</i>. https://www.nist.gov/quantum-information-science/quantum-computing-explained IBM. (2025). <i>What is quantum computing?</i> IBM Thinking. https://www.ibm.com/think/topics/quantum-computing.				

Session 2: Operating System and Software Installation

1.C	2.C	3.A	4.C	5.B
6. 32-bit: Is capable of less memory (to 4 GB RAM). 64-bit: Supports newer processors and is able to work with more memory.				
7. Restart the computer. Check for software updates. Reinstall the application if necessary or needed. Check for conflicting programs or missing files.				
8.To identify and thwart viruses, malware and ransomware that can damage files or steal data.				
9. Administrative privileges give a user permission to install, modify, or remove software and system settings.				
10. GO to Control Panel → click on Programs → than click on Programs and Features. Select the program to remove or right click . Click Uninstall and follow the prompts. Restart the PC if required.				

Research Task:

Research Task:

Artificial intelligence is becoming directly integrated into the contemporary operating systems, and one of the most significant releases is Microsoft Windows 11 Copilot. Copilot is a built-in AI assistant which assists users to conduct daily tasks, including summarising documents, creating emails, and making settings changes or locating information without the need to open several applications. It is developed by using cloud-based AI services provided by Microsoft and integrates Bing Chat and GPT-based language models into the operating system. To the staff of IT support, this trend would imply that they will soon be dealing with an environment where the operating system is automating users assistance and configuration. Although this can minimize the number of help-desk calls that happen on a regular basis, it also comes at a price of control over data privacy, the difficulty in accessing AI in a secure manner, and the problem of troubleshooting erratic AI reactions. With the increasing number of vendors integrating AI in operating systems, including Siri intelligence in the Apple operating system and open-source AI systems in different Linux operating systems, technicians will be required to learn the advantages and shortcomings of the systems. AI-based operating systems have the potential to streamline the process of support in the future, yet they will also require new competencies in handling automation and managing data in an ethical way.

References

Microsoft. (2025). *Meet Windows Copilot in Windows 11*. <https://www.microsoft.com/windows/windows-11-copilot>

ZDNet. (2024). *Windows 11 Copilot: How AI is changing the desktop experience*. <https://www.zdnet.com/article/windows-11-copilot-explained>

Session 3: Network Fundamentals and Connectivity Troubleshooting

1.B	2.B	3.B	4.C	5.B
6.Public IP: It is employed to define the devices in the internet; globally distinct. Private IP: It is utilised internally in a local network; it cannot be transmitted on the internet.				
7.The default gateway or DNS server is not configured or is down.				
8. The cable is over damaged physically or not completely plugged. The switch port or connector RJ45 is faulty.				
9.The ping command is used to test accuracy of connection between devices, by executing packets and measuring the response time. It aids in establishing whether a network route or equipment is accessible.				
10. MAC Address: A unique hardware identifier assigned to a network interface card (NIC). IP Address: A logical address assigned by the network for communication. <i>Difference:</i> MAC = physical and permanent; IP = logical and changeable.				

Research Task:

The most recent wireless networking technology is Wi-Fi 6E that extends the existing Wi-Fi 6 technology up to the frequency band of 6 GHz. This new range offers a lot of bandwidth and a reduced congestion and speed up to 9.6 Gbps when the ideal conditions are applied. It enables more devices to be connected at the same time and with reduced latency, which makes it suitable to the modern workplaces that require cloud services, video conferencing, and Internet of Things (IoT) devices. Wi-Fi 6E has new opportunities and tasks to cover by the professionals in tech support. The support staff should know how to set up routers and network adapters compatible with the 6 GHz band, compatibility with older devices, and interference problems. Since Wi-Fi 6E can support the high-density environments effectively, it eliminates numerous problems related to the connectivity that are frequent in the shared office or school environment. It also implies, however, that helpdesk technicians need to be informed of firmware updates, driver management, and country-specific regulatory specifications. Altogether, Wi-Fi 6E is a major leap towards a fast and more stable wireless network that can allow users and IT professionals to experience a more streamlined performance and necessitate continuous learning of network management and security.

References

Wi-Fi Alliance. (2024). *Wi-Fi 6E: The next generation of connectivity*. <https://www.wi-fi.org/discover-wi-fi/wi-fi-6e>

Cisco. (2025). *Understanding Wi-Fi 6 and 6E for enterprise networks*. <https://www.cisco.com/c/en/us/solutions/enterprise-networks/wi-fi-6e.html>

Session 4: System Maintenance, Troubleshooting and Ticket Management

1.B	2.B	3.B	4.B	5.C
6. Too many startup programs or background processes. Insufficient RAM or low disk space. Malware or outdated system software.				
7. It identifies and eliminates malicious applications capable of slowing down the system or consuming too many resources.				
8. Soft Reboot: Restarting the system using the operating system (e.g., Start → Restart). Used for updates or minor issues. Hard Reboot: Physically holding the power button to force shutdown. Used when the system is frozen or unresponsive.				

9. Take notes regarding any troubleshooting performed.

Status of the ticket should be changed to Escalated or the relevant technician of higher level assigned to it.

Inform the user that the problem has been reported and he or she will be contacted to help him out.

10. It HELPS other technicians/supervisors to know about the actions, root cause, and the ultimate solution.

Stabilizes proper records, transparency and better troubleshooting in future.

Research Task:

Artificial Intelligence (AI) is changing the way IT support teams in the modern world handle and solve user-related problems. It is now possible to see AI-based chatbots respond to frequent questions in real time, troubleshoot with users, and even automatically generate service tickets. Machine learning can enable support systems to process large datasets to forecast failures, also called predictive maintenance, to enable technicians to resolve issues before users lose working time. These applications enhance productivity, minimize waiting time, and set human agents to do more challenging issues. Nevertheless, there are also downsides of AI in the area of tech support. Chatbots may lack a clear understanding of the intent of the user and therefore become a source of frustration or unresolved problems. Predictive systems need connection to sensitive information, which is of concern to privacy and security. Also, AI models require continual updates and monitoring to be precise and impartial. To the IT professionals, knowledge on how to combine and manage AI tools will be an essential skill. With the advancement of AI technology, the question of balance between automation and human care will determine the future of customer service, a mix of machine and human factor of a professional technician.

References

IBM. (2025). *How AI is transforming IT support operations*. <https://www.ibm.com/topics/ai-it-support>

Gartner. (2024). *AI in service management: Opportunities and challenges*. <https://www.gartner.com/en/articles/ai-in-service-management>

Session 5: Advanced Troubleshooting and Hardware Diagnostics

1.A	2.C	3.C	4.D	5.B
6.Check memory errors with the Windows Memory Diagnostic tool or the MemTest86. It is also possible to test by taking one of the RAM sticks out at a time in order to isolate the faulty module.				
7.Microscopic cracks between the heat sink and the CPU are filled with thermal paste to enhance the transfer of heat and to avoid overheating of the processor.				
8.There is a probability that the hard drive is in failure (the click of death). The advice here is to make sure that all the valuable data is saved at the earliest and as much as possible, the hard drive should be changed.				
9.It may appear that a slow computer has poor RAM or CPU, however, malware, excessive start up programs or corrupt operating system are the actual problem.				
10. It assists in organisation, documentation and troubleshooting. Technicians are able to locate parts and connections much faster eliminating confusion and human error in the process of repairs.				
Research Task: The computer industry is also being transformed by ARM processors which introduces the efficiency of the mobile chip design to laptops and desktops. In comparison to the conventional Intel or AMD x86 processors, the ARM processors have a simplified instruction set that consumes less power and yet, is very high-performing. This is an architecture that was initially found in smartphones, but is presently available in products like Apple M-series of Macbooks, and other Windows laptops with Qualcomm Snapdragon processors. To the IT support professionals, this change presents an opportunity as well as a challenge. ARM systems have increased battery life, less noisy cooling, as well as reduced boot time, which enhance user experience. But compatibility is also a consideration: certain older Windows programs that are built to run on x86 processors might not run directly on ARM chips as both will have to be emulated or software replacements made. Support technicians should know how to deal with cross-platform performance problem, firmware upgrades, and image the system of various types of CPUs. With additional manufactures using ARM, tech support departments will have to be trained on hybrid environments comprising of x86 and ARM devices. This trend in the long term is expected to provide more efficient and portable computing, although it will require flexible IT professionals with the ability to deal with architectures that are different. References Apple. (2024). <i>Apple silicon: The future of Mac</i>. https://www.apple.com/mac/m1/ Qualcomm. (2025). <i>Snapdragon X Elite: Bringing ARM power to Windows laptops</i>. https://www.qualcomm.com				

Session 6: Advanced Network and Final Troubleshooting Scenario

1.B	2.B	3.B	4.B	5.B
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6. First we need to Set the hostname and domain name
Then Configure IP management address and default gateway.

7. Incorrect folder permissions or sharing settings.
The **network discovery** or **file sharing** feature may be turned off.

8. To make sure that the issue is fully resolved and that no new problems were introduced or found.
Then Confirms that the user's functionality is restored before closing the ticket.

9.
Listen actively and remain calm.
Acknowledge their frustration and apologize for the inconvenience.
Clearly explain what steps you'll take, keep them updated, and resolve the issue promptly.

10. Verify with the user that the issue is resolved.
Test all systems involved.

Document the troubleshooting steps and resolution summary.
Change the ticket status to "**Solved**" or "**Closed**".

Research Task: The position of IT support professional will change radically over the next ten years as technology will be more automated, cloud-like, and integrated with artificial intelligence. Three key areas of skills, such as automation and AI management, cybersecurity awareness, and effective communication skills, will have to be developed by the technicians to remain effective.

To start with, with automation and AI-based systems replacing regular troubleshooting, support techs will have to be trained on how to monitor, train, and optimize these tools. Knowing the way AI-generated reports should be interpreted and complex issues that cannot be addressed by automation should be escalated will be essential.

Second, cybersecurity will be one of the priorities. As the number of security threats increases (phishing, ransomware, data breaches, etc.), all IT support workers need to know the basics of security practices, multi-factor authentication, and data management security. Tech support will not only repair computers, but also enable the protection of organizations against cyber risks. Lastly, soft skills such as communication, empathy and problem-solving will take on a new value like never before. With the increasing smartness of technology, humans will continue to require help by humans to know the solutions and feel at ease. Clear-cut presentation of technical problems, stress management, and establishing trust with users will make professionals stand out.

Technical care will become a combination of technical skills, security obligation and human warmth in future to make sure that people as well as technology are going to collaborate with each other in a safe and efficient manner.

References

CompTIA. (2024). *The future of IT support: Human skills meet automation*. <https://www.comptia.org>
Gartner. (2025). *Top skills for the next decade of tech support*. <https://www.gartner.com>

#5 Computer won't start – no power

Submitted November 5, 2025 at 18:17 **Received via** Web Form **Requester** omar khan <omarkhan.00@gmail.com>

CCs
Tirth Patel <tirthp052@gmail.com>

Status category	Ticket status	Type	Priority	Group	Assignee
Solved	Solved	Incident	High	Support	Tirth Patel

Summary

The customer reported that their desktop computer will not start and shows no power. IT Support acknowledged the issue and initiated troubleshooting, promising to provide an update soon. No further actions or resolutions have been documented yet.

Summary date and time	Summary agent ID	Summary locale	Language	Sentiment	Language confidence	Sentiment confidence
2025-11-05T05:17:39Z	14241647951759	en-US	English	Neutral	High	High

Tirth Patel November 5, 2025 at 18:17

Hi Omar,

Thank you for contacting IT Support.
I understand your desktop computer is not powering on. I'll begin troubleshooting this issue and update you shortly.

Kind regards,
Tirth Patel
IT Support

Tirth Patel November 5, 2025 at 18:21

Internal note

Troubleshooting Log:

1. Checked power cable – OK.
 2. Checked PSU switch – ON.
 3. Opened case using ESD strap.
 4. Found 24-pin motherboard power connector loose.
 5. Repeated connector – system powered on.
- Resolution: Loose connection on motherboard. Computer now working.

Support Software by Zendesk

#6 New Software Installation Request

Submitted November 5, 2025 at 18:28 **Received via** Web Form **Requester** omar khan <omarkhan.00@gmail.com>

Status category	Ticket status	Type	Due date	Priority	Group	Assignee
Solved	Solved	Task		Normal	Support	Tirth Patel

Intent	Intent confidence	Language	Sentiment	Language confidence	Sentiment confidence
Software::Software connection::Connection issue	High	English	Neutral	High	High

Tirth Patel November 5, 2025 at 18:28

Hi Omar,

Thank you for contacting IT Support. I understand you'd like an office suite, media player, and antivirus software installed on your new computer. I'll proceed with installing free, reliable versions of each and will update you once the setup is complete.

Kind regards,
Tirth patel
IT Support

Tirth Patel November 5, 2025 at 18:30

Hi Omar,

All requested software has been successfully installed on your computer:

- LibreOffice – Office suite (Word, Excel, PowerPoint alternatives)
→ Open from Start Menu > LibreOffice.
- VLC Media Player – For video and audio playback
→ Open from Start Menu > VLC Media Player.
- AVG Free Antivirus – For malware and virus protection
→ Icon available in system tray (bottom right).

Each program was tested and is running correctly.
Please restart your computer if prompted and confirm everything works as expected.

Kind regards,
Tirth Patel
IT Support

Support Software by Zendesk

#7 Training Room Network Setup for 10 Students

Submitted
November 5, 2025 at 18:36

Received via
Web Form

Requester
Visiting Instructor <xyz1234@gmail.com>

Status category
Solved

Ticket status
Solved

Type
Task

Priority
Normal

Group
Support

Assignee
Tirth Patel

Intent
Software::Software connection::Connection issue

Intent confidence
Low

Language
English

Sentiment
Neutral

Language confidence
High

Sentiment confidence
High

Tirth Patel November 5, 2025 at 18:36
Network Setup – Ticket 3N-NW-001

internal note

1. Connected Cisco switch to instructor's network jack (Port 1).
2. Connected 10 student PCs to switch ports 2–11.
3. Connected instructor PC to port 12.
4. Verified all connections and link lights on switch.
5. Used ipconfig command on each computer to confirm IP configuration:
 - IP range: 192.168.10.2 – 192.168.10.12
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.10.1
6. Verified internet access on all student PCs (ping google.com successful).
7. Created shared folder "TrainingShare" on instructor PC (C:\TrainingShare).
8. Enabled "Everyone" with Read/Write permissions.
9. Accessed shared folder from Student PC (\\InstructorPCName\TrainingShare) – read/write successful.
10. Documented configuration and confirmed all systems working.

Result: Network fully operational; all student PCs have internet and shared folder access.

Tirth Patel November 5, 2025 at 18:37

Hi [xyz],

The training room network has been successfully configured.
All 10 student computers have internet access and can access the shared folder on your computer.

- Details:
- Network connected via Cisco switch.
 - Shared folder: \\InstructorPCName\TrainingShare
 - Access: Students can read and write files.

Everything has been tested and confirmed operational.
Please let me know if you'd like any additional software or restrictions added.

Kind regards,
Tirth
IT Support

Support Software by Zendesk

#9 Computer running slow and freezing

Submitted
November 5, 2025 at 18:41

Received via
Web Form

Requester
omar khan <omarkhan.00@gmail.com>

Status category
Solved

Ticket status
Solved

Type
Incident

Priority
Normal

Group
Support

Assignee
Tirth Patel

Intent
Misc::Unsolicited::Spam

Intent confidence
Low

Language
English

Sentiment
Neutral

Language confidence
High

Sentiment confidence
High

Tirth Patel November 5, 2025 at 18:41

Hi Omar,

Thank you for contacting IT Support. I understand your computer has been running slowly and freezing frequently.
I'll perform a full system maintenance check to improve performance and stability. I'll update you once the process is complete.

Kind regards,
TIRTH
IT Support

Tirth Patel November 5, 2025 at 18:42

Hi Omar,

I have completed system maintenance on your computer to resolve the slow performance and freezing issues.

- Actions taken:
- Cleaned up temporary files and emptied Recycle Bin
 - Disabled unnecessary startup programs
 - Performed full antivirus scan (no malware detected)
 - Defragmented the hard drive for better speed
 - Installed the latest Windows updates

After these optimizations, your system is running smoothly and no longer freezing.
Please monitor it and let us know if the issue reappears.

Kind regards,

IT Support

Support Software by Zendesk

#10 Explained Power Supply Failure to End User

Submitted November 5, 2025 at 18:46		Received via Web Form		Requester omar khan <omarkhan.00@gmail.com>	
Status category Solved	Ticket status Solved	Type Task	Priority Low	Group Support	Assignee Tirth Patel
Intent Order::Product issue::Product not working as expected		Intent confidence Low	Language English	Sentiment Neutral	Language confidence High
				Sentiment confidence Medium	
Tirth Patel November 5, 2025 at 18:46 Professional Communication Exercise – Activity 5.1					
1. Issue: Desktop computer not powering on due to failing PSU. 2. Technician explained issue in simple, user-friendly terms: - Described what the power supply does. - Explained why it failed (wear and tear). - Outlined replacement process and time required. 3. Asked user for consent to proceed with replacement. 4. User approved the repair. 5. Replaced PSU and tested system – working fine. 6. User satisfied with explanation and repair outcome.					
Tirth Patel November 5, 2025 at 18:46 Hi Omar, During our session, we discussed the issue with your desktop computer not turning on properly. I explained that the problem was caused by a faulty power supply — the internal part that delivers power to all other components. I also explained that replacing the power supply is a straightforward process and asked for your approval to proceed. You agreed to have the repair completed. The power supply has now been replaced, and your system is functioning normally. Kind regards, IT Support					
Support Software by Zendesk					

Submitted November 5, 2025 at 18:58		Received via Web Form		Requester omar khan <omarkhan.00@gmail.com>	
Status category Solved	Ticket status Solved	Type Incident	Priority Normal	Group Support	Assignee Tirth Patel
Intent Software::Software connection::Connection issue		Intent confidence Medium	Language English	Sentiment Neutral	Language confidence High
				Sentiment confidence High	
Tirth Patel November 5, 2025 at 18:58 Intermittent Network Issue – Ticket 6N-NW-001					
1. Physical Checks: - Verified Ethernet cable connection on both ends. - Replaced cable with a tested working one – issue persisted intermittently. - Checked Cisco switch port – link light flickering. - Moved connection to a different port on the switch – link became stable. - Verified NIC (network card) properly seated. 1. Command-Line Diagnostics: - Ran 'ping google.com -t' for 5 minutes – observed packet loss (~30%). - Checked IP settings using 'ipconfig /all' – found DHCP lease expiring frequently. - Renewed IP address using 'ipconfig /release' and 'ipconfig /renew'. - Checked active connections using 'netstat -a' – no suspicious or duplicate connections. 1. Resolution: - Fault traced to intermittent Cisco switch port (original port 6). - Reassigned computer to port 10 on switch. - Flushed DNS and renewed IP lease. - Confirmed stable connectivity with 0% packet loss after 10-minute ping test. 1. Verification: - Internet connection stable. - Access to shared folder confirmed. - No further disconnections observed.					
Tirth Patel November 5, 2025 at 18:59 Hi Omar, Thank you for reporting the issue with your computer losing network connectivity at times. I'll begin by testing the physical and network settings to identify the cause and will update you once I've completed troubleshooting. Kind regards, IT Support					
Tirth Patel November 5, 2025 at 18:59 Hi Omar, The issue was caused by an intermittent network port on the switch. I moved your connection to a new port and refreshed your network settings. Your computer now has a stable connection and can access the internet and shared drives without interruptions. Please monitor and let me know if you experience further drops. Kind regards, IT Support					
Support Software by Zendesk					

